

### 33. Morphological operations based on OpenCV using Morphological Gradient technique.

#### AIM:

Morphological operations using Morphological Gradient technique.

#### PROGRAM:

```
import cv2
import numpy as np

img = cv2.imread(
r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png",
cv2.IMREAD_GRAYSCALE
)

if img is None:
    print("Error: Image not found!") else:
    # Create a 3x3 kernel    kernel =
    np.ones((3, 3), np.uint8)

    # Apply Morphological Gradient    gradient = cv2.morphologyEx(img,
cv2.MORPH_GRADIENT, kernel)

    # Save the output image    cv2.imwrite(
r"C:\Users\Susmitha\Downloads\ITA0510 Lab\Morphological_Gradient.jpg",
gradient
    )

    # Display the images    cv2.imshow("Original
Image", img)    cv2.imshow("Morphological
Gradient", gradient)

    cv2.waitKey(0)
cv2.destroyAllWindows() INPUT:
```



**OUTPUT:**



**RESULT:**

Thus the program is executed successfully.